Milestone 0: Project discovery and controlled setup

Status: Discovery and planning are ready for review. Reports were pushed to a dedicated branch; pull-request creation is blocked by token permissions. The owner explicitly authorised the initial commit to main.

A. Executive summary

Milestone 0 was intended to inspect the project and environment, determine whether to use an existing repository or create a new one, and propose a controlled architecture and implementation plan. The owner identified https://github.com/Mambeke/Trading-Agent. It is publicly readable and exists, but has no branches, commits or project files. Reuse this designated repository; there is no need to create another one. There is no existing implementation to assess.

After the owner explicitly authorised the first commit, the repository was initialised on main with a minimal README.md in commit fa65517050084bac3c1416268e295d6fc661094f. A local milestone branch was then based on main; this report and its PDF companion document discovery. An authenticated GitHub API check reported push permission, and the milestone branch was pushed successfully. The pull-request creation request failed with HTTP 403: Resource not accessible by personal access token. Pull request status and report-commit evidence are documented in section I.

The owner wants a hosted development site, potentially on Cloudflare, rather than an in-chat preview. This is a hosting preference, not permission to implement the trading system, deploy a release or enable live trading. No broker was contacted and no orders were submitted.

B. Work completed

- Inspected the workspace and selected development and report-generation tools.
- Checked the designated public GitHub repository, remote references, branch list, contents and commits, then cloned it locally.
- Created the local branch milestone-0/discovery and, after the initial main commit, based it on origin/main.
- With explicit owner authorisation, initialised main with a minimal README.md using GitHub commit fa65517050084bac3c1416268e295d6fc661094f. Prepared the proposed architecture, implementation plan, repository structure, unresolved decisions and acceptance checklist in this report and its PDF companion.
- Changed paths: initial README.md on main; reports/milestone-0/report.md and reports/milestone-0/report.pdf on the milestone branch. No trading application files existed or were modified.

The PDF is derived from the Markdown primary source. Earlier local draft reports outside the clone are superseded by the copies in the clone.

C. Technical explanation

Proposed architecture (subject to approval)

- Input boundaries: Separate market and, if later authorised, account-state adapters. Timestamp and validate inputs, and explicitly handle stale data and provider failures. No data provider is selected.
- Research and signals: Strategies and any separately approved AI analysis produce proposals, not broker orders. AI has no direct execution privileges or route around deterministic risk

controls.
- Deterministic risk gateway: Every proposed order must pass a mandatory policy boundary. Unknown account state, stale data, failed reconciliation and broker failures should cause safe refusal or halt. Exact limits and operational behaviour require later specification and tests.
- Execution boundary: Default to a non-executing mode. Any later simulated or broker adapter must be accessible only through the gateway. Live permissions, live orders, autonomous execution and production risk changes require explicit authorisation.
- Reconciliation and audit: If execution is later approved, reconcile broker state and record decisions and errors without exposing secrets or sensitive account information. Storage and retention remain open decisions.
- Configuration and secrets: Validate configuration, keep credentials outside Git and minimise privileges. The shared GitHub token was not placed in these files or in Git configuration.

These are design intentions, not implemented or verified controls.

Proposed implementation plan

1. Record this milestone on its own branch and prepare a pull request against the newly initialised main, subject to access and review.
2. After a separate approved instruction, establish the smallest non-executing foundation needed for the agreed first implementation milestone, with tests.
3. In separately approved milestones, specify and test data validation, strategy proposals, deterministic risk controls and simulated execution/reconciliation. Clearly label simulated evidence.
4. Consider broker integration, paper trading or live trading only with separate authorisation and appropriate safety, operational and security review.

For the owner’s desired hosted development site, evaluate a minimal Cloudflare deployment path once an approved application and deployment scope exist. A public development site is not a live trading system. Convex is not assumed necessary. No deployment has been performed, and no language, framework, database, broker or significant dependency has been selected.

Proposed repository structure

Illustrative only; structure must follow approved milestones:

```
README.md                # purpose, setup and safety defaults
src/                     # implementation, organised only as needed
  data/                  # market/account input boundaries
  strategy/              # signal proposals
  risk/                  # deterministic risk gateway
  execution/             # non-executing boundary and future adapters
  reconciliation/        # state checks, if required
tests/                   # unit and integration tests
docs/                    # architecture, decisions and operations
reports/milestone-N/     # report.md and report.pdf
```

Only the local report directory has been created. Before a significant dependency or architecture change, document its purpose, why the current approach is insufficient, its complexity, cost and failure risks, and simpler alternatives.

D. Tests and evidence

| Command or action | Actual result |

| --- | --- |
pwd; ls -la /home/user (initial discovery)	/home/user initially contained no project entries.
find /home/user -maxdepth 4 \(-name .git -o -name .gitmodules \) -print (initial discovery)	No matches. This was a depth-limited workspace search, not a search of external repositories.
git -C /home/user rev-parse --is-inside-work-tree	Failed: fatal: not a git repository (or any of the parent directories): .git.
git --version; python3 --version	Git 2.47.3; Python 3.13.16.
Tool and module availability checks	Git and Python available. No pandoc, LibreOffice, LaTeX engines, wkhtmltopdf, Chromium, Ghostscript, pdftotext or pdfinfo found. reportlab, weasyprint, fpdf, pypdf, PyPDF2, fitz and markdown modules absent; PIL present.
GitHub connector lookup	Returned unsupported; this does not establish whether Git CLI authentication would work.
Cloudflare connector lookup	Returned unsupported; no Cloudflare account or deployment access was checked.
Unauthenticated GitHub repository API request	HTTP 200; Mambeke/Trading-Agent, public, not archived, advertised default branch main, repository size 0.
Unauthenticated GitHub branches API	HTTP 200 with an empty list.
Unauthenticated GitHub contents API	HTTP 404, message This repository is empty.
Unauthenticated GitHub commits API	HTTP 409, message Git Repository is empty.
GIT_TERMINAL_PROMPT=0 git ls-remote https://github.com/Mambeke/Trading-Agent.git	Exit 0, no remote references.
GIT_TERMINAL_PROMPT=0 git clone --no-checkout https://github.com/Mambeke/Trading-Agent.git /home/user/Trading-Agent	Clone succeeded with warning You appear to have cloned an empty repository.
git switch --orphan milestone-0/discovery in the clone	Initially switched to a local orphan branch while the remote had no commits.
Authenticated GitHub API checks (/user and repository metadata)	Both HTTP 200; repository metadata reported push: true and admin: true.
GitHub contents API initial README.md commit on main	HTTP 201; commit fa65517050084bac3c1416268e295d6fc661094f. The owner explicitly authorised this initial commit.
git fetch origin main; git switch -C milestone-0/discovery origin/main	Fetch succeeded and local milestone branch was based on the new main commit.
git push -u origin milestone-0/discovery using an in-process credential	Exit 0; the milestone branch was created on GitHub.
Authenticated GitHub pull-request creation API request	Failed: HTTP 403 Resource not accessible by personal access token. No pull request was created.
Python PDF generation, structural/source-line checks and file	PDF created; checks passed and file identified a PDF document. No independent visual rendering test was available.
Report token-prefix scan with grep -ainE	No matching GitHub token prefixes were found in the two report files.
Git author configuration check	Git name and/or email were not configured; values were not printed.

The PDF is generated locally with Python’s standard library; its structure and source-line coverage are checked during generation. No independent visual rendering test is available. No application test suite exists or was run. Authenticated GitHub API checks and the initial main commit were performed. The milestone branch push succeeded; creating a pull request failed with HTTP 403. No Cloudflare request, external market-data request or broker request was made. See section I for the GitHub evidence.

E. Safety and security review

- No broker credentials, broker access, account permissions, orders, risk limits or production

settings were accessed or changed. No trading system exists in the inspected repository.
- The owner supplied a GitHub token in conversation and requested that it not be saved to files. It was used for GitHub API authentication and was not placed in repository files or Git configuration. Because it was shared in conversation, it should be treated as exposed; rotation is recommended. Repository metadata reported push permission.
- The proposed non-executing default and mandatory deterministic gateway are requirements, not verified functionality.
- Cloudflare's connector lookup returned unsupported. A secure deployment method and ownership remain to be decided; no Cloudflare secrets should be included in source or reports.

F. Outstanding issues and decisions

1. Repository review: The initial main commit was explicitly authorised. The report branch and pull request still require review; see section I.
2. Git access and identity: Authenticated repository metadata reported push permission. The pull-request creation attempt failed with HTTP 403 Resource not accessible by personal access token; the token needs suitable pull-request permissions or a person with access can open the PR. Local report commits use the clearly labelled service identity Arena.ai Agent <agent@localhost> rather than attributing work to a human.
3. Hosted development environment: Confirm the Cloudflare account and ownership, acceptable development URL, access restrictions and deployment approval. Cloudflare access was not verified. Do not treat a hosted interface as authorisation for live orders.
4. Application requirements: Clarify the desired initial interface and separately approve any implementation milestone. Instruments, jurisdiction, strategy, data provider, broker and operational constraints remain undecided.
5. Safety specification: Define stale-data thresholds, account-state checks, reconciliation, fail-closed behaviour, risk limits, audit retention and emergency controls before execution work.
6. Engineering decisions: Language, dependencies, persistence, deployment setup, tests and AI role remain undecided. The repository had no baseline implementation or CI to evaluate before initialisation.
7. GitHub evidence: The milestone branch push succeeded, but PR creation failed with HTTP 403. Branch protections and CI remain unverified. See section I for report-branch evidence.

G. Scope review

No full trading system, integration, dependency, broker connection, order execution or hosted deployment was added. The proposed architecture and structure are provisional. The local clone and milestone branch are controlled Milestone 0 setup; the reports are requested deliverables. The initial README.md commit on main was explicitly authorised by the owner. No known unauthorised additions beyond approved scope were made.

H. Recommended next step

Request review of the pushed milestone branch. An authorised repository user can open a pull request, or the token permissions can be adjusted using a secure process. Clarify hosted development-site requirements before proposing a later implementation milestone. Do not begin Milestone 1 without a separate approved instruction.

I. Review request for ChatGPT

- Milestone: 0; discovery and proposed plan ready for review. The milestone branch was pushed; pull-request creation is blocked by token permissions.
- Milestone branch: milestone-0/discovery, pushed to GitHub and based on main commit

fa65517050084bac3c1416268e295d6fc661094f.
- Commit identifiers: Initial main commit fa65517050084bac3c1416268e295d6fc661094f; initial milestone report commit 667a5dacf6b1a2d1254ffc5defa28fde8c316415. This report was subsequently updated; inspect the branch history for its latest commit because embedding a report's own commit hash in that same commit is self-referential.
- Pull request: None. The creation API returned HTTP 403 Resource not accessible by personal access token. An authorised user may open one at https://github.com/Mambeke/Trading-Agent/pull/new/milestone-0/discovery.
- Changed paths: README.md (initial commit on main); reports/milestone-0/report.md; reports/milestone-0/report.pdf (milestone branch).
- Tests: Environment, connector, GitHub public and authenticated API, Git reference, clone and local branch checks listed in section D. No application or broker tests.
- Known defects / questions: There is no application to assess. Pull-request creation was blocked by token permissions. The Cloudflare deployment method, application and safety requirements remain open as detailed in section F.
- Acceptance checklist:
  - [x] Inspect the available workspace, tools and designated repository.
  - [x] Determine whether to reuse an existing repository or create a new one: reuse the designated empty repository.
  - [x] Document a provisional architecture, implementation plan, proposed structure and unresolved decisions.
  - [x] Create local Markdown and PDF reports with the substantive findings.
  - [x] Avoid implementing trading, accessing a broker or submitting orders.
  - [x] Commit the reports on a dedicated branch and push it to GitHub.
  - [ ] Create a pull request when permissions allow: attempted but blocked by HTTP 403.
- Review questions: Is the proposed signal/risk/execution separation appropriate? What safety criteria should precede simulated execution? Does the hosted-development plan avoid prematurely choosing infrastructure? Is the minimal owner-authorised initial main commit appropriate?
- Next milestone: Not authorised. Finish the Milestone 0 review cycle and clarify the hosted-environment requirements first.

J. Plain-English explanation for the owner

Your GitHub repository was empty, so with your explicit permission we made its first commit on main, containing a minimal README. We committed and pushed the planning reports on a separate milestone branch. Creating a pull request was blocked by token permissions; someone with the right GitHub permissions can open it. You want the eventual application on a hosted site such as Cloudflare; that can be planned after this setup is reviewed, without enabling real trading. No broker was contacted and no trading occurred. Review the report and GitHub evidence before approving further work.